WORLD HISTORY

MODULE - 16

AGRICULTURAL AND INDUSTRIAL REVOLUTIONS - 1

AGRICULTURAL REVOLUTION

- The term agricultural revolution refers to the radical changes in the method of agriculture in England in the 17th and 18th centuries.
- There was a massive increase in agricultural productivity, which supported the growing population.
- The Agricultural Revolution preceded the Industrial Revolution in England.

During the Agricultural Revolution, four key changes took place in agricultural practices..

1	• Enclosure of lands
2	• Mechanization of farming
3	• Four-field crop rotation
4	• Selective breeding of domestic animals.

AGRICULTURAL REVOLUTION

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1 Prior to the agricultural revolution, the practice of agriculture had been much the same across Europe since the Middle Ages.

2 The open field system was essentially feudal. Each farmer engaged in cultivation in common land and dividing the produce.

3 From the beginning of 12th century, some of the common fields in Britain were enclosed into individually owned fields.

4 This process rapidly accelerated in the 15th and 16th centuries as sheep farming grew more profitable. This led to farmers losing their land and their grazing rights. Many farmers became unemployed.

5 In the 16th and 17th centuries, the practice of enclosure was denounced by the Church, and legislation was drawn up against it.

AGRICULTURAL REVOLUTION

Learning Space

6 However, the mechanization of agriculture during the 18th century required large, enclosed fields. This led to a series of government acts, culminating finally in the General Enclosure Act of 1801.

7 By the end of the 19th century the process of enclosure was largely complete. Great experiments were conducted in farming during this period.

8 Machines were introduced for seeding and harvesting. Rotation of crops was introduced by Townshend. The lands became fertile by this method.

9 Bakewell introduced scientific breeding of farm animals. The horse-drawn ploughs, rake, portable threshers, manure spreaders, multiple ploughs and dairy appliances had revolutionized farming.

10 These changes in agriculture increased food production as well as other farm outputs.

INDUSTRIAL REVOLUTION

1 The term 'Industrial Revolution' was used by European scholars – Georges Michelet in France and Friedrich Engels in Germany.

2 It was used to describe the changes that occurred in the industrial development of England between 1760 and 1820.

3 The Industrial Revolution had far-reaching effects in England.

4 Subsequently, similar changes occurred in European countries and in the U.S.A. The Industrial Revolution had a major impact on the society and economy of these countries and also on the rest of the world.

5 The primary cause of the Industrial Revolution was the scientific inventions.

Learning Space

INDUSTRIAL REVOLUTION

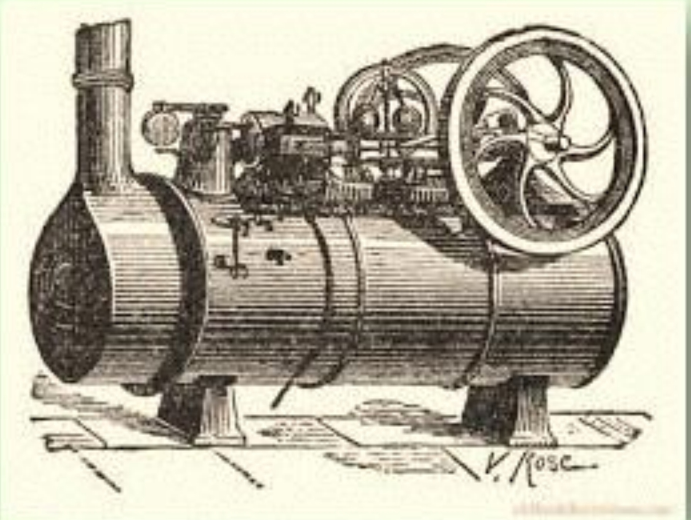
6 This phase of industrial development in England is strongly associated with new machinery and technologies. These made it possible to produce goods on a massive scale compared to handicraft and handloom industries.

7 There were changes in the cotton and iron industries.

8 Steam, a new source of power, began to be used on a wide scale in British industries.

9 Its use led to faster forms of transportation by ships and railways.

10 Industrialization led to greater prosperity for some, but in the initial stages many people including women and children had experienced poor living and working conditions. This sparked off protests and the government was forced to enact laws to improve the conditions of workers.



CAUSES OF INDUSTRIAL REVOLUTION IN ENGLAND

REASONS

1. England's advantageous geographical location.
2. The precedence of agricultural revolution.
3. New inventions and the introduction of machinery.
4. The enterprising spirit of British entrepreneurs.
5. Growth of capital in England.
6. Colonial possessions of England, which supplied raw materials and served as markets.

TEXTILE MACHINERY

- The earliest mechanical inventions came in the textile industry.
- Spinning was the slowest process in the manufacturing of cloth.
- The invention of flying shuttle by Kay in 1733 improved weaving.
- In 1764, Hargreaves invented the 'spinning jenny'. This machine could spin eight threads at the same time, instead of one.
- Arkwright improved the 'spinning jenny' in 1769.
- Compton improved it still further in 1779.

SCIENTIFIC INVENTIONS

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TEXTILE MACHINERY

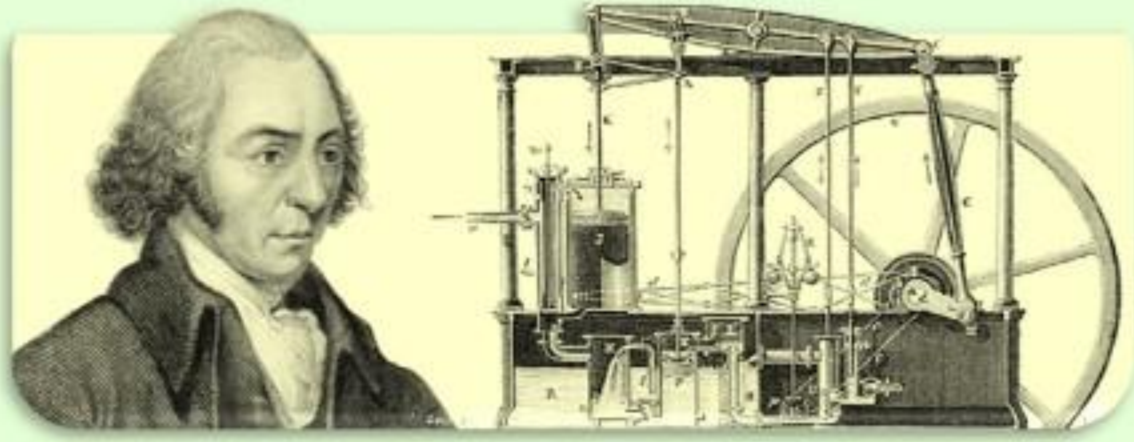
- In 1785, Cartwright invented the Power loom.
- Whitney, an American, speeded up the process (1792) with a Cotton gin, which automatically removed seeds from the fiber of the cotton.
- The invention of the sewing machine by Elias Howe, in 1846, accelerated the production of clothing and made possible the modern clothing industry.
- Thus, one invention followed another, not only in textile industries but also in many others. In this way, the present-day complex machinery has evolved.

Edmund Cartwright

Eli Whitney

STEAM ENGINE

- Heavy machinery could not function without power to operate it.
- The invention of the Steam engine provided the practical solution.
- The first practical application of steam to machinery was made by James Watt in 1765.
- He devised the first closed cylinder with a piston pushed back and forth by steam. This has been extensively used in textile machinery.



DEVELOPMENT OF TRANSPORT

There is a close relationship between the development of industry and improvement in transportation. Industrialization depends largely on the bringing of raw materials to factories and on the disposing of manufactured goods in a wide market.

ROADWAYS

- ✓ As late as the 17th century, highways were poorly kept.
- ✓ A pack horse was the only possible means of travel on land.
- ✓ In the second half of the 18th century, John McAdam (1756-1836) built a type of hard-surfaced road in England.
- ✓ The only important change made in this method was the substitution of a tar composition for mud as a binder.
- ✓ France copied the English methods, and under the patronage of the government many highways were built.
- ✓ The heavy expenses involved in the building and upkeep of highway encouraged the development of inland waterways.

DEVELOPMENT OF TRANSPORT

INLAND WATERWAYS

- During the second half of the 18th century and the early part of the 19th century thousands of miles of artificial water route were dug in England, in France, and in the United States.
- In 1761, a canal was built in England from Worsley to Manchester to carry coal from the mines to the furnaces.
- There were serious drawbacks in the river and canal transportation.
- The rate of travel was slow and the expense of construction and maintenance was high.
- Geographical factors limited the extent to which water transportation could be utilized.

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DEVELOPMENT OF TRANSPORT

RAILWAYS

- ✓ Railroads provided a solution for these problems.
- ✓ The first tracks were made of wood and the first cars were horse drawn, but the introduction of iron for rails and the application of Watt's steam engine for traction power revolutionized the whole procedure.
- ✓ George Stephenson constructed the first practical locomotive in 1814.
- ✓ The Stockton and Darlington railroad started operation in England in 1825.
- ✓ The era of railroads had begun.

COMMUNICATION

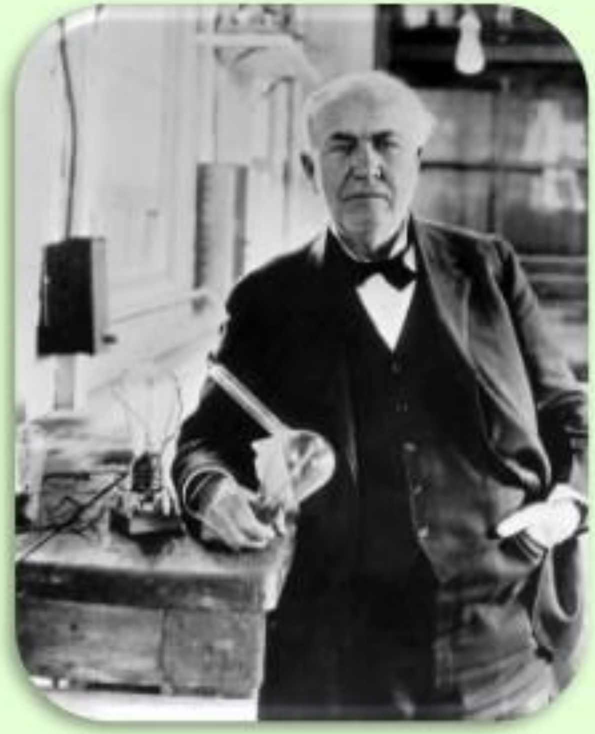
- Modern transportation and business enterprises are much dependent on rapid and efficient communication.
- Before the perfection of the telegraph, carrier pigeons and semaphores were the speediest methods available.
- The electric telegraph depended upon earlier basic researches made by Faraday, Volta, Ampere, and Franklin.
- It was invented independently in Germany, England, and the United States, by Steinheil, Wheatstone, and Morse, respectively.

COMMUNICATION

- Telegraphic equipment was widely installed after 1845.
- A cable from America to Europe was laid under the Atlantic Ocean in 1866.
- By the close of the 19th century, all the important commercial centers in the world had telegraphic communications.
- The penny post was established in 1840.
- The Universal Postal Union, to aid international mail service, was adopted in 1875.
- Graham Bell invented the telephone in 1876.

LIGHTING

- In industry, transportation, social activities, amusements, and cultural pursuits, artificial light plays a very important role.
- In 1784, a burner was devised for oil lamps, which was later used for kerosene lamps.
- Gas for artificial illumination was introduced and widely used by the middle of the 19th century.
- Davy, in 1821, worked out the theory of the electric arc.
- Edison, in 1879, invented the electric bulb.



IRON AND STEEL

- The coal and iron industries replaced old technologies of wood, water and wind.
- In 1709 Darby introduced coal for charcoal in blast furnace.
- John Smeaton invented the blast furnace with a rotary fan.
- Henry Cord and Peter Onions introduced puddling and rolling Process in 1784.
- In 1740 steel was produced at Sheffield by Huntsman.
- Later, Henry Bessemer invented a faster and cheaper method of producing steel.
- The first iron bridge was constructed in 1777.
- The first iron ship was made in 1790.

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suggestions@learningspace.in

To Contact us:
info@learningspace.in

For Information:
info.learningspacedigital@gmail.com

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0866 244 44 72
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